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TREND-BASED MOCK PAPER

GATE Life Sciences

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Life Sciences
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section B: Life Sciences

Q1. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q2. [1-NAT] [1 mark]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q3. [1-NAT] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q5. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q6. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q7. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q8. [1-NAT] [1 mark]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-NAT] [1 mark]

$d/dx[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

(Numerical Answer Type — Enter the numerical value)

Q10. [1-MCQ] [1 mark]

The differential equation $y dx - x dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q11. [1-NAT] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q13. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q14. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q15. [2-NAT] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q17. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
 - (B) Continuous but not differentiable at 0
 - (C) Neither continuous nor differentiable
 - (D) Differentiable at 0 only
-

Q18. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
 - (B) e^2
 - (C) 1
 - (D) infinity
-

Q19. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
 - (B) $2\pi r$
 - (C) πr
 - (D) $2\pi r^2$
-

Q20. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
 - (B) $\|u\| + \|v\|$
 - (C) $\|u\| / \|v\|$
 - (D) $\sqrt{(\|u\| * \|v\|)}$
-

Q21. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q22. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both

(D) Neither

Q23. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q24. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q25. [2-NAT] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q27. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q28. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q30. [2-MCQ] [2 marks]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q31. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
 - (B) 0
 - (C) -1
 - (D) $\cos(\pi)$
-

Q33. [1-NAT] [1 mark]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-NAT] [1 mark]

pH of 10^{-8} M HCl approx:

(Numerical Answer Type — Enter the numerical value)

Q35. [1-MCQ] [1 mark]

Galvanic cell converts:

- (A) Electrical to chemical
 - (B) Chemical to electrical
 - (C) Heat to work
 - (D) Mechanical to electrical
-

Q36. [1-NAT] [1 mark]

First-order half-life with $k=0.693 \text{ min}^{-1}$:

(Numerical Answer Type — Enter the numerical value)

Q37. [1-NAT] [1 mark]

$K_p = K_c(RT)^{\Delta n}$. For $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$, $\Delta n =$

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MSQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
 - (B) 7
 - (C) 8
 - (D) 6.98
-

Q39. [2-NAT] [2 marks]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MSQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8

(D) 6.98

Q41. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q42. [1-MCQ] [1 mark]

SN2 rate depends on:

- (A) Substrate only
- (B) Substrate + nucleophile
- (C) Nucleophile only
- (D) Solvent

Q43. [1-NAT] [1 mark]

Benzene undergoes:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q45. [1-NAT] [1 mark]

Grignard reacts with:

(Numerical Answer Type — Enter the numerical value)

Q46. [1-NAT] [1 mark]

Geometrical isomerism in:

(Numerical Answer Type — Enter the numerical value)

Q47. [2-MCQ] [2 marks]

Benzene undergoes:

- (A) Only addition
- (B) Only substitution
- (C) Both
- (D) Elimination

Q48. [2-NAT] [2 marks]

Grignard reacts with:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-MCQ] [2 marks]

Benzene undergoes:

- (A) Only addition
- (B) Only substitution

- (C) Both
- (D) Elimination

Q50. [2-MCQ] [2 marks]

Hofmann elimination gives:

- (A) Most substituted alkene
- (B) Least substituted
- (C) Mixture
- (D) No product

Q51. [1-MCQ] [1 mark]

Van der Waals corrects for:

- (A) Only volume
- (B) Only pressure
- (C) Both volume and pressure
- (D) Temperature

Q52. [1-NAT] [1 mark]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

(Numerical Answer Type — Enter the numerical value)

Q53. [1-NAT] [1 mark]

pH of 0.01M HCl:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-NAT] [1 mark]

$\Delta G = -5\text{kJ/mol}$ at 300K. Equilibrium constant:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

First-order: 50% in 10 min. 90% time:

- (A) 10 min
- (B) 33.2 min
- (C) 20 min
- (D) 5 min

Q56. [2-NAT] [2 marks]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

pH of 0.01M HCl:

- (A) 1
- (B) 2
- (C) 0.01
- (D) 14

Q58. [2-MCQ] [2 marks]

Rate doubles per 10K rise. Activation energy ($T=300\text{K}$):

- (A) $\sim 110\text{ kJ/mol}$

- (B) ~ 56 kJ/mol
- (C) ~ 22 kJ/mol
- (D) ~ 8 kJ/mol

Q59. [1-MCQ] [1 mark]

Werner theory explains:

- (A) Bonding in complexes
- (B) CN and geometry
- (C) Magnetic properties
- (D) Color only

Q60. [1-MCQ] [1 mark]

Strongest field ligand (spectrochemical series):

- (A) I⁻
- (B) Br⁻
- (C) CN⁻
- (D) F⁻

Q61. [1-NAT] [1 mark]

Octahedral crystal field splits d-orbitals into:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

[Co(NH₃)₆]³⁺ coordination number:

- (A) 3
- (B) 4
- (C) 6
- (D) 8

Q63. [2-MCQ] [2 marks]

[Fe(CN)₆]⁴⁻ (low spin, d₆) is:

- (A) Diamagnetic
- (B) Paramagnetic
- (C) Ferromagnetic
- (D) Antiferromagnetic

Q64. [2-NAT] [2 marks]

Werner theory explains:

(Numerical Answer Type — Enter the numerical value)

Q65. [2-MCQ] [2 marks]

Octahedral crystal field splits d-orbitals into:

- (A) t_{2g} and e_g
 - (B) e_g and t_{2g} only
 - (C) s and p
 - (D) sigma and pi
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	D	1
2	1MCQ	A	1
3	1NAT	A	1
4	1MCQ	B	1
5	1MCQ	C	1
6	1NAT	B	1
7	1MCQ	C	1
8	1NAT	B	1
9	1NAT	D	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1MCQ	C	1

14	1NAT	A	1
15	1MCQ	A	1
16	1NAT	B	1
17	1NAT	A	1
18	1NAT	B	1
19	1NAT	B	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1NAT	C	1
23	1NAT	C	1
24	1NAT	A	1
25	1NAT	B	1
26	1MCQ	B	1
27	1MCQ	B	1
28	1NAT	B	1

29	1MCQ	C	1
30	1NAT	B	1
31	2MCQ	A	2
32	2MCQ	C	2
33	2MCQ	A	2
34	2NAT	C	2
35	2NAT	A	2
36	2MCQ	A	2
37	2MCQ	B	2
38	2MCQ	B	2
39	2MCQ	A	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2MCQ	A	2
43	2MCQ	A	2

44	2MCQ	A	2
45	2NAT	B	2
46	2MSQ	D	2
47	2NAT	D	2
48	2MCQ	A	2
49	2NAT	B	2
50	2MCQ	B	2
51	2NAT	A	2
52	2MCQ	C	2
53	2MCQ	A	2
54	2MCQ	B	2
55	2NAT	B	2
56	2NAT	B	2
57	2MSQ	D	2
58	2MCQ	B	2

59	2MCQ	B	2
60	2NAT	B	2
61	2MCQ	B	2
62	2MCQ	A	2
63	2MCQ	C	2
64	2MCQ	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q2.

Answer: A

Solution:

Homogeneous ODE. Substituting $y = x \cdot v$ and $y(1) = 0$ gives $v = 0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q3.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q4.

Answer: B

Solution:

SN₂: bimolecular. Rate = $k[\text{substrate}][\text{nucleophile}]$. Both affect rate.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: difficult

Q5.

Answer: C

Solution:

$(P + a/V^2)(V - b) = RT$. Corrects for intermolecular forces (a) and finite volume (b).

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q6.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q7.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q8.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q9.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q10.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q11.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q12.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q13.

Answer: C

Solution:

CN⁻ (strongest) > NH₃ > en > NO₂⁻ > F⁻ > Cl⁻ > Br⁻ > I⁻ (weakest).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q14.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: d_{x²-y²}, d_{z²}).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q15.

Answer: A

Solution:

$L\{t^s f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310))$. $E_a = R \ln(2)/(1/300 - 1/310) = 53.6$ kJ/mol. Closest: 56.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q17.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1$ min.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q18.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q19.

Answer: B

Solution:

$\Delta_n = 2 - 4 = -2$.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q20.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q22.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q23.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

$\Delta G = -RT \ln(K) \Rightarrow K = \exp(-\Delta G/RT) = \exp(5000/(8.314*300)) = e^2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: easy

Q25.

Answer: B

Solution:

Grignard: reacts with CO_2 (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q26.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: B

Solution:

$t_{90} = t_{50} * \log(10)/\log(2) = 10 * 3.32 = 33.2 \text{ min}$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q28.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2,1,0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: C

Solution:

6 NH_3 ligands: CN = 6. Octahedral geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: moderate

Q30.

Answer: B

Solution:

Geometrical: restricted rotation around double bond. 2-butene has cis/trans.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q31.

Answer: A

Solution:

Low-spin d_6 : t_{2g}^6 , all paired. Diamagnetic.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q32.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q34.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q35.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q36.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q37.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q38.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q40.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

Row 2 = $2 \times$ Row 1, Row 3 = $3 \times$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

$$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

$$\text{PGF of Poisson}(\lambda): G(z) = E[z^X] = \exp(\lambda(z-1)).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

$$f(0)=2, f(1)=0, f(-1)=4, f(2)=4. f(-2)=-8+6+2=0. \text{ Roots at } x=1 \text{ (double?) and } x=-2. f(x)=(x-1)^2(x+2). \text{ Two distinct real roots.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}} \approx 1.1e-7$. pH ≈ 6.96 .

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q47.

Answer: D

Solution:

Perchloric acid ($HClO_4$) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: difficult

Q48.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1) \approx 0.68$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

$$2 = \exp(E_a/R \cdot (1/300 - 1/310)). E_a = R \ln(2) / (1/300 - 1/310) = 53.6 \text{ kJ/mol. Closest: 56.}$$

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: difficult

Q50.

Answer: B

Solution:

$$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q52.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{ Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q55.

Answer: B

Solution:

Grignard: reacts with CO₂ (carboxylic acid), carbonyls, epoxides. Reacts with protic solvents.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: moderate

Q56.

Answer: B

Solution:

Werner: primary valency (oxidation state) + secondary valency (CN). Explained geometry.

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q57.

Answer: D

Solution:

Very dilute: [H⁺] from water dominates. [H⁺]_{total} approx 1.1e-7. pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q58.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q59.

Answer: B

Solution:

Benzene: aromatic stability favors electrophilic aromatic substitution.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: difficult

Q60.

Answer: B

Solution:

$(1 + 1/n)^{2n} = [(1 + 1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: B

Solution:

Hofmann (bulky base): gives least substituted (Hofmann) alkene.

Subject: Organic Chemistry | Topic: Organic Chemistry | Difficulty: easy

Q62.

Answer: A

Solution:

Octahedral: t_{2g} (lower: d_{xy}, d_{yz}, d_{zx}) and e_g (higher: d_{x²-y²}, d_{z²}).

Subject: Inorganic Chemistry | Topic: Inorganic Chemistry | Difficulty: easy

Q63.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q64.

Answer: B

Solution:

$[H^+] = 10^{-2}$. $pH = -\log(10^{-2}) = 2$.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

Q65.

Answer: B

Solution:

$2 = \exp(E_a/R \cdot (1/300 - 1/310))$. $E_a = R \cdot \ln(2) / (1/300 - 1/310) = 53.6 \text{ kJ/mol}$. Closest: 56.

Subject: Physical Chemistry | Topic: Physical Chemistry | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 53 marks
Chemistry: 9 questions, 13 marks
Organic Chemistry: 9 questions, 13 marks
Physical Chemistry: 8 questions, 11 marks
Inorganic Chemistry: 7 questions, 10 marks

Difficulty Distribution:

Easy: 18 | Moderate: 35 | Difficult: 12

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.